

User Manual

for S32K1_S32M24X WDG Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	07.03.2025	NXP RTD Team	S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
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This User Manual describes the NXP Semiconductor AUTOSAR Watchdog (Wdg) for the **S32K1_S32M24X**. AUTOSAR Wdg driver configuration parameters and deviations from the specification are described in the Driver chapter of this document. The requirements and APIs of the AUTOSAR Wdg driver are described in the AUTOSAR Wdg Driver Software Specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116_qfn32
- s32k116_lqfp48
- s32k118_lqfp48
- s32k118_lqfp64
- s32k142_lqfp48
- s32k142_lqfp64
- s32k142_lqfp100
- s32k142w_lqfp48
- s32k142w_lqfp64
- s32k144_lqfp48
- s32k144_lqfp64 / MWCT1014S_lqfp64
- s32k144_lqfp100 / MWCT1014S_lqfp100
- s32k144_mapbga100
- s32k144w_lqfp48
- s32k144w_lqfp64

- s32k146_lqfp64
- s32k146_lqfp100 / MWCT1015S_lqfp100
- s32k146_mapbga100 / MWCT1015S_mapbga100
- s32k146_lqfp144
- s32k148_lqfp100
- s32k148_mapbga100 / MWCT1016S_mapbga100
- s32k148_lqfp144
- s32k148_lqfp176
- s32m241_lqfp64
- s32m242_lqfp64
- s32m243_lqfp64
- s32m244_lqfp64

All of the above microcontroller devices are collectively named as S32K1_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language
EWM	External Watchdog Monitor
WDOG	Watchdog Timer
AEWDOG	Alive watchdog

2.5 Reference List

#	Title	Version
1	Specification of Driver	AUTOSAR Release R21-11
2	S32K1xx Reference Manual	Rev.14.1, 01/2024
3	S32M24x Reference Manual	Rev. 5, 12/2024
4	S32K1xx Data Sheet	Rev. 14, 08/2021
5	S32M2xx Data Sheet	Rev. 7, 12/2024
6	S32K116 Errata	Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024
7	S32K118 Errata	Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024
8	S32K142 Errata	Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024
9	S32K144 Errata	Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024
10	S32K144W Errata	Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024
11	S32K146 Errata	Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024
12	S32K148 Errata	Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024
13	S32M242 Errata	Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024
14	S32M244 Errata	Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)
- [Usage in non AUTOSAR context](#)

3.1 Requirements

Requirements for this driver are detailed in the Autosar Driver Software Specification document (See [Table Reference List](#)).

3.2 Driver Design Summary

The Watchdog Timer(WDOG) and External watchdog monitor(EWM) with programmable interrupt response are available in S32M244:

The Watchdog Timer (WDOG) is a peripheral module that can prevent system lockup in situations such as software getting trapped in a loop or if a bus transaction fails to terminate. When enabled, the WDOG requires periodic execution of a watchdog servicing operation. The servicing operation resets the timer to a specified time-out period. If this servicing action does not occur before the timer expires the WDOG generates an interrupt or hardware reset. The WDOG can be configured to generate a reset or interrupt on an initial time-out. However the WDOG reset is always generated after RCM delay in LPO cycles depending on configuring the RCM_SRIE[DELAY] and RCM_SRIE[WDOG] bits.

In addition to these two modes of operation, the watchdog timer also supports a windowed mode. In this mode, the service sequence must be performed in the last part of the time-out period defined by the window register. The window is open when the down counter is less than the value in the WDOG_WIN register. Outside of this window, service sequence writes that the Watchdog will reset the MCU.

The WDOG has the following features:

- 16-bit time-out register to set the time-out period

- Programmable selection of window mode or regular servicing
- Programmable selection of reset or interrupt on an initial time-out
- Programmable selection of test mode or user mode

The External Watchdog Monitor (EWM) is designed to monitor external circuits, as well as the microcontroller software flow. This provides a back-up mechanism to the internal watchdog that resets the microcontroller's CPU and peripherals. The overflow of the watchdog counter must not occur if the software code works well and services the watchdog to re-start the actual counter. The EWM differs from the internal watchdog in that it does not reset the microcontroller's CPU and peripherals. The EWM provides an independent EWM_OUT_b signal that when asserted resets or places an external circuit into a safe mode. The EWM_OUT_b signal is asserted upon the EWM counter time-out. An optional external input EWM_in is provided to allow additional control of the assertion of EWM_OUT_b signal actual counter.

The EWM has the following features:

- 8-bit time-out register to set the time-out period
- Programmable selection of window mode or regular servicing
- Programmable selection of reset or interrupt on an initial time-out
- Programmable selection of the logic level voltage of the input pin if enabled and when is asserted

3.3 Hardware Resources

The WDG driver uses the WDOG, EWM, and AEWDG hardware IPs.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR WDG Driver software specification in some places. The table below identifies the AUTOSAR requirements that are not implemented or out of scope for the WDG Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently or out of scope for the WDG driver.

Requirement	Status	Description	Notes
SWS_Wdg_00055	N/S	The Wdg module for an external watchdog driver shall have source code that is independent of the microcontroller platform.	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately

Requirement	Status	Description	Notes
SWS_Wdg_00034	N/S	The start address of the watchdog trigger routine shall be statically configurable to a fixed memory location by the user. The user needs to take care that Configured memory location is valid for the platform on which driver is being implemented on. This configuration parameter shall only be given if supported/needed by the hardware.	Rejection Reason: N/A for any of the available platforms
SWS_Wdg_00076	N/S	To access the external watchdog hardware, the corresponding Wdg module instance shall use the functionality and API of the corresponding handler or driver, e.g. the SPI handler or DIO driver.	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately
SWS_Wdg_00162	N/S	The routine servicing an external watchdog shall be implemented by usage of an own internal hardware timer to be independent from other peripherals or by using a GPT driver callback	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately
SWS_Wdg_00077	N/S	A Wdg module for an external watchdog shall satisfy the same functional requirements and offer the same functional scope as a Wdg module for an internal watchdog. Hence their respective APIs are semantically identical.	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately
SWS_Wdg_00078	N/S	The Wdg module shall add all parameters required for accessing the external watchdog hardware, e.g. the used SPI channel or DIO port, to the module's published parameters and to the module's configuration parameters.	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately
SWS_Wdg_00172	N/S	If more than one watchdog driver instance exists on an ECU (namely an external and an internal one) the A↔PI names and instance specific type names specified in this chapter shall be made unique by expansion according to SRS_BSW_00347.	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg R↔TD module. External watchdog must be supported separately
SWS_Wdg_00175	N/S	These requirements are not applicable to this specification.	This requirement defines a list of requirements marked in the standard as Not applicable

Requirement	Status	Description	Notes
ECUC_Wdg_00118	N/S	Name - WdgTriggerLocation - Parent Container - WdgGeneral - Description - Location (memory address) of the watchdog trigger routine. - Multiplicity - 1 - Type - EcucFunction - NameDef - Default value - - - max - Length - - - minLength - - - regular - Expression - - - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - - - - Post-build time - - - - Scope / Dependency - scope: localdependency: Only relevant if provided by hardware and needed by the system. -	This parameter functionality is replaced by CPR_RTD_00161.wdg.
ECUC_Wdg_00112	N/S	Container Name - WdgExternal - Configuration - Description - Configuration items for an external watchdog hardware - Configuration Parameters -	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg RTD module. External watchdog must be supported separately
ECUC_Wdg_00113	N/S	Name - WdgExternalContainerRef - Parent Container - WdgExternal - Configuration - Description - Reference to either a DioChannelGroup container in case the hardware watchdog is connected via DIO pins or a SpiSequenceConfiguration container in case the watchdog hardware is accessed via SPI - Multiplicity - 0..1 - Type - Choice reference to [DioChannelGroup , SpiSequence] - Post-Build Variant Multiplicity - true - Post-Build Variant Value - true - Multiplicity Configuration Class - Pre-compile time - X - VARIANT-Pre-Compile - Link time - X - VARIANT-Link-Time - Post-build time - X - VARIANT-Post-Build - ILD - Value Configuration Class - Pre-compile time - X - VARIANT-Pre-Compile - Link time - X - VARIANT-Link-Time - Post-build time - X - VARIANT-Post-Build - LD - Scope / Dependency - scope: localdependency: See DIO resp. SPI SWS -	This requirement is rejected because it refers to External Watchdog modules, which are not in scope for the Wdg RTD module. External watchdog must be supported separately

3.5 Driver Limitations

- AeWdog is not support notification is not implemented by Wdg driver.
- To allow generating multiple instances from same project, the Wdg Tresos component has parallel generation

disabled (via setting `canRunParallel = false` in file `plugin.xml`), which leads to increased generation time.

3.6 Driver usage and configuration tips

The Wdg driver can function in either Direct Service Mode or Gpt Triggered mode. In Direct Service Mode, the Wdog and Ewm peripherals can be serviced directly, while in Gpt Triggered mode, a callback notification is set up so that the Gpt will periodically trigger the Wdog and Ewm peripherals.

1. Configure the Wdg reference clock from Mcu (see parameter `WdgClkSrcRef`) according to reference point used by the Wdg peripherals on the platform.
2. In Gpt Triggered Mode, configure Wdg routine used for triggering as a Gpt callback (`Wdg_Cbk_Gpt`↔`NotificationX` must be configured as a notification callback for the Gpt channel intended for triggering).
3. The interrupt feature of the Wdog IP (and Wdg Instance 0) can be used to delay the moment until the system is reset, but can not prevent or stop it. This allows some time to prepare the application for the incoming reset. More details on how to configure the Wdog Interrupt feature can be found in the Reference Manual.
4. The Wdg driver can be configured to run from RAM or ROM targets from the `WdgRunArea` option. The RAM target should be used in case the module dependencies are also run from RAM, such in the case of bootloaders. If the module dependencies code is being run from a flash target, the ROM option needs to be used.
5. When configured for window mode, the Wdg can be reset if it is serviced outside of the window period.
6. In Direct Service Mode, the Gpt module does not need to be configured and the user must call `Wdg_43_↔InstanceX_Service` periodically to service the Wdg in order to avoid the system is reset. The `Wdg_Instance↔X_SetTriggerCondition` function is unavailable in this mode. Direct Service Mode can not be used in parallel with Gpt Triggered mode.
7. When called for WDG instance over EWM peripheral, the function `Wdg_ChannelSetMode` raises DET (if enabled) and returns error, because EWM peripheral only supports Slow Mode.
8. In Gpt Triggered Mode, the user must ensure the following conditions are met when servicing the Wdg:
 - Watchdog timeout period and Trigger Condition must not have similar values because of interrupt clashing.
 - Watchdog timeout period must be less than Trigger Condition, in order for the Wdg to be serviced in time.
 - To avoid potential clashing of triggering interrupts with the main loop, the user should avoid choosing a hardware trigger that is equal to or equal to multiples of the Servicing Period.
 - Trigger Condition must be greater than the Servicing period with which `Wdg_SetTriggerCondition` is called, in order for the Wdg not to expire.

3.7 Runtime errors

The driver generates the following DEM errors at runtime.

Function	Error Code	Condition triggering the error
<code>Wdg_43_Instance<Number>_Init</code>	<code>WDG_E_DISABLE_REJECTED</code>	Initialization or mode switch failed because it would disable the watchdog has occurred
<code>Wdg_43_Instance<Number>_↔SetMode</code>	<code>WDG_E_DISABLE_REJECTED</code>	Initialization or mode switch failed because it would disable the watchdog has occurred

Function	Error Code	Condition triggering the error
Wdg_43_Instance<Number>_Init	WDG_E_MODE_FAILED	Setting a watchdog mode failed (during initialization or mode switch) has occurred
Wdg_43_Instance<Number>_↔ SetMode	WDG_E_MODE_FAILED	Setting a watchdog mode failed (during initialization or mode switch) has occurred

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like: `#define <Mip>Conf_<Container_ShortName>_<Container_ID>`

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

3.9 Usage in non AUTOSAR context

For the non-ASR environments, both High-Level and IP layer interfaces are exposed and can be used by software units above. These may consist of either higher level NXP SW deliverables, like stacks and middleware, or the application code itself.

The goal of the product is to be versatile, easy to use and integrate with other SW environments, offering both standardized and optimized low-level interfaces.

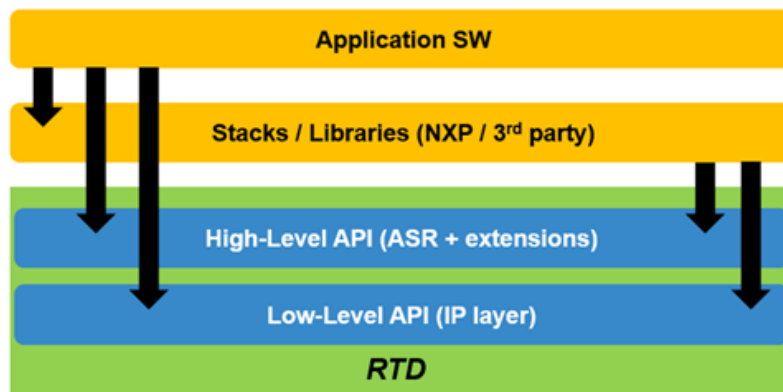


Figure 3.1 RTD Usage in non-ASR Context

In the non-AUTOSAR environment, there are no specific ECU states defined, but an external assumption for the application is that the correct sequence of initialization takes place prior to calling any runtime APIs.

The constraints considered for defining the APIs in the non-AUTOSAR environment are:

- Hardware resources used by the drivers must be properly initialized prior to runtime usage by calling the initialization function with configuration parameters generated by the configuration tools
- All dependencies must be initialized prior to the usage of a driver that depends on them; this translates to a general initialization sequence applicable for all applications, as follows:
 1. Initialization of the device clock tree so other periphery can be initialized
 2. Configuration of the pinout so peripherals with external connections can communicate with the outside world

3. Configuration of the common resources (e.g. DMA channels, memory regions, access rights, etc) prior to the usage of any runtime functionality that has to benefit from this overall SoC configuration
4. Initialization of specific drivers
5. Runtime functionality
6. Deinitialization (inverse initialization order)

High-level and IP layer functionalities cannot be used simultaneously over the same hardware resource. The granularity of the resource must be defined specifically for each driver, but as a general rule the default granularity considered is the peripheral instance (e.g. Gpt high level APIs cannot be called interchangeably with Pit_Ip low level APIs over the same PIT hardware instance).

The diagram below shows a generic, high-level sequence of calls to RTD interfaces in the non-AUTOSAR scenario:

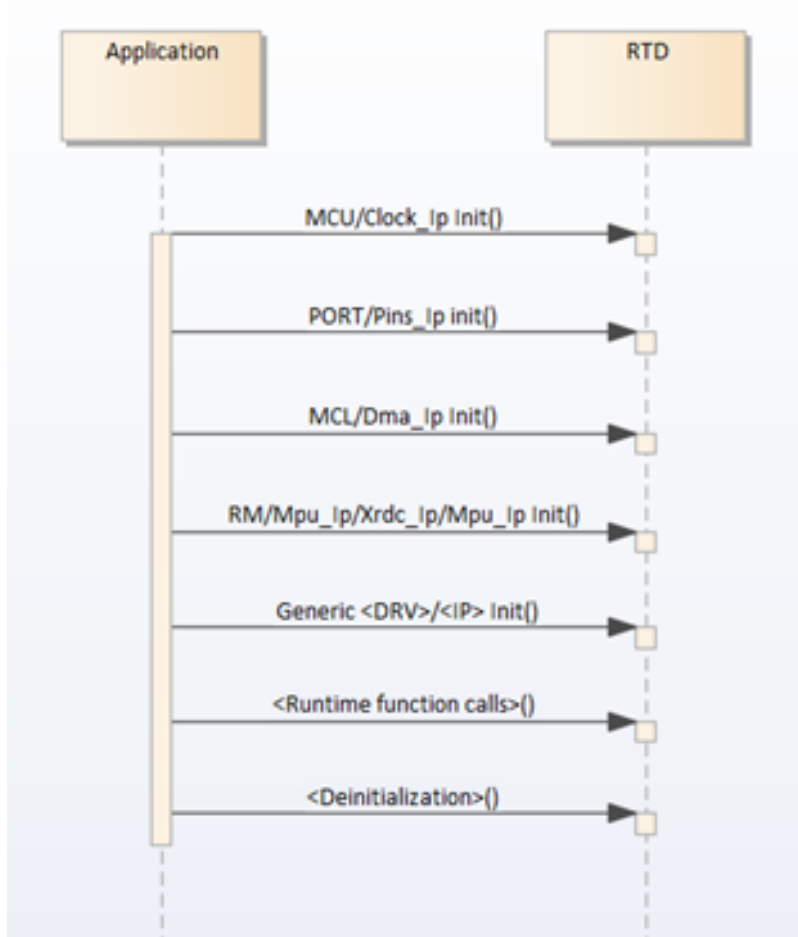


Figure 3.2 Non-AUTOSAR Sequence diagram

Chapter 4

Configuration Parameters

RTD provides configuration schema for the high level driver in EB Tresos proprietary format (**.xdm**), S32CT proprietary format (**.component**), but also in a standardized AUTOSAR format (**.epd**). For the low level driver configuration only the S32CT proprietary format (**.component**) is supported. The configuration data of the project can be saved in EB Tresos proprietary format (**.xdm**) and AUTOSAR standardized format (**.epc**) from EB Tresos and in S32CT proprietary format (**.mex**) and AUTOSAR standardized format (**.epc**) from S32 Design Studio. RTD provides templates for parsing the configuration data and generating configuration structures from it in XPath format, to be used with the EB Tresos generator and in JS format, to be used with the S32CT generator. Since the generation process is not standardized by AUTOSAR, only the two mentioned tools can be used with the provided templates. Using the same input configuration data, the same configuration structures will be obtained using any of the generators.

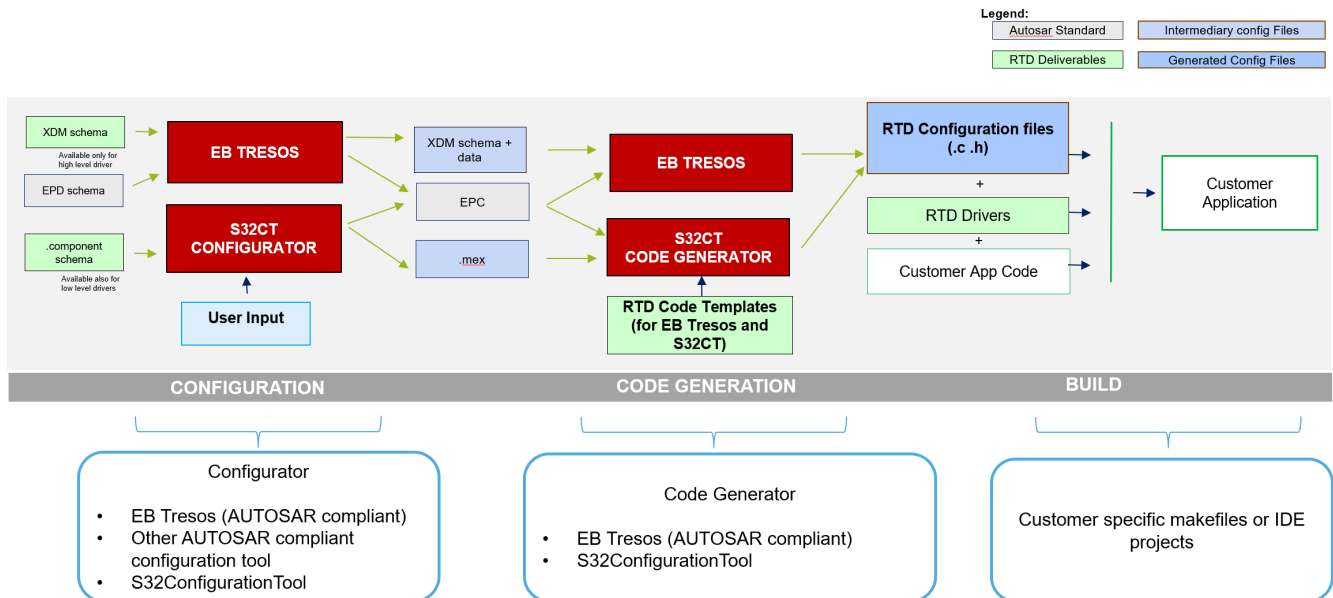


Figure 4.1 Configuration generation flow

- Wdg
- WDOG_IP
- EWM_IP
- AEWDOG_IP

4.1 Wdg

- Module [Wdg](#)
 - Container [WdgDemEventParameterRefs](#)
 - * Reference [WDG_E_DISABLE_REJECTED](#)
 - * Reference [WDG_E_MODE_FAILED](#)
 - Container [WdgGeneral](#)
 - * Parameter [WdgDevErrorDetect](#)
 - * Parameter [WdgDisableAllowed](#)
 - * Parameter [WdgEnableUserModeSupport](#)
 - * Parameter [WdgEnableDirectService](#)
 - * Parameter [WdgEnableMultiPartitionSupport](#)
 - * Parameter [WdgTimeoutMethod](#)
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 - * Parameter [WdgCallbackNotification](#)
 - * Parameter [WdgVersionInfoApi](#)
 - * Reference [WdgEcucPartitionRef](#)
 - * Container [AutosarExt](#)
 - Parameter [WdgDisableDemReportErrorStatus](#)
 - Container [WdgClockReferencePoint](#)
 - * Reference [WdgClockReference](#)
 - Container [WdgSettingsConfig](#)
 - * Parameter [WdgInstance](#)
 - * Parameter [WdgDefaultMode](#)
 - * Parameter [WdgInterruptContentEnable](#)
 - * Reference [WdgExternalTriggerCounterRef](#)
 - * Container [WdgExternalConfiguration](#)
 - Reference [WdgExternalContainerRef](#)
 - * Container [WdgSettingsFast](#)
 - Parameter [WdgClockValue](#)
 - Parameter [WdgRunsInStopMode](#)
 - Parameter [WdgRunsInDebugMode](#)
 - Parameter [WdgRunsInWaitMode](#)

- Parameter [WdgOperationMode](#)
- Parameter [WdgClockSelection](#)
- Parameter [WdgTimeoutPeriod](#)
- Parameter [WdgWindowMode](#)
- Parameter [WdgWindowPeriod](#)
- Parameter [WdgPrescalerEnabled](#)
- Parameter [WdgAllowUpdates](#)
- Reference [WdgClkSrcRef](#)
- * Container [WdgSettingsOff](#)
 - Parameter [WdgAllowUpdates](#)
- * Container [WdgSettingsSlow](#)
 - Parameter [WdgClockValue](#)
 - Parameter [WdgRunsInStopMode](#)
 - Parameter [WdgRunsInDebugMode](#)
 - Parameter [WdgRunsInWaitMode](#)
 - Parameter [WdgOperationMode](#)
 - Parameter [WdgClockSelection](#)
 - Parameter [WdgTimeoutPeriod](#)
 - Parameter [WdgWindowMode](#)
 - Parameter [WdgWindowPeriod](#)
 - Parameter [WdgPrescalerEnabled](#)
 - Parameter [WdgAllowUpdates](#)
 - Reference [WdgClkSrcRef](#)
- Container [WdgPublishedInformation](#)
 - * Parameter [WdgTriggerMode](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1.1 Module Wdg

Wdg

Configuration of the Wdg (Watchdog driver) module

Included containers:

- [WdgDemEventParameterRefs](#)
- [WdgGeneral](#)
- [WdgClockReferencePoint](#)
- [WdgSettingsConfig](#)
- [WdgPublishedInformation](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME, VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.1.2 Container WdgDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.3 Reference WDG_E_DISABLE_REJECTED

Reference to the DemEventParameter which shall be issued when the error "Initialization or mode switch failed because it would disable the watchdog" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.4 Reference WDG_E_MODE_FAILED

Reference to the *DemEventParameter* which shall be issued when the error "Setting a watchdog mode failed (during initialization or mode switch)" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.5 Container WdgGeneral

WdgGeneral

All general parameters of the watchdog driver are collected here.

Included subcontainers:

- [AutosarExt](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.6 Parameter WdgDevErrorDetect

Wdg Development Error Detect

Compile switch to enable / disable development error detection for this module.

This switch enables the Development error detection for the individual hardware IP associated with the Wdg Instance.

If at least one Wdg Instance has this switch enabled, the High Level code of the Wdg Driver will have the development error detection feature enabled for all enabled Wdg Instances.

True: Development error detection enabled

False: Development error detection disabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.7 Parameter WdgDisableAllowed

Wdg Disable Allowed

Compile switch to allow / forbid disabling the watchdog driver during runtime.

True: Disabling the watchdog driver at runtime is allowed

False: Disabling the watchdog driver at runtime is not allowed

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.8 Parameter WdgEnableUserModeSupport

Wdg Enable User Mode Support

When this parameter is enabled, the Wdg module will adapt to run from User Mode, with the following measure : configuring REG_PROT for Wdg IPs so that the registers under protection can be accessed from user mode by setting UAA bit in REG_PROT_GCR to 1 for more information and availability on this platform, please see chapter "User Mode Support" in IM.

True: Wdg module will adapt to run from User Mode.

False:Wdg module will not adapt to run from User Mode.

If this parameter is not ediatable, that means Wdg driver can run in User Mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.9 Parameter WdgEnableDirectService

Wdg Enable Direct Service

When this parameter is enabled, the Wdg module can be serviced directly, without using an external hardware trigger.

True: Wdg module can be serviced directly.

False:Wdg module can not be serviced directly.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Configuration Parameters

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.10 Parameter WdgEnableMultiPartitionSupport

Wdg Enable Multi-Partition Support

When this parameter is enabled, the Wdg module will adapt to run in Multi-Partition

True: Wdg module will adapt to run in Multi-Partition.

False:Wdg module will not adapt to in Multi-Partition.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.11 Parameter WdgTimeoutMethod

WdgTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If OSIF_COUNTER_SYSTEM or OSIF_COUNTER_CUSTOM are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_DUMMY', 'OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM']

4.1.12 Parameter WdgOsifTimeoutVal

Wdg Osif timeout value.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	3000
max	4294967295
min	0

4.1.13 Parameter WdgIndex

Wdg Instance 0 Index

Represents the watchdog driver's ID for Instance 0 so that it can be referenced by the watchdog interface.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.1.14 Parameter WdgInitialTimeout

Wdg Initial Timeout

The initial timeout (sec) for the trigger condition to be initialized during Init function. It shall be not larger than WdgMaxTimeout.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	65.535
min	0.0

4.1.15 Parameter WdgMaxTimeout

Wdg Max Timeout

The maximum timeout (sec) to which the watchdog trigger condition can be initialized.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	65.535
min	0.0

4.1.16 Parameter WdgRunArea

Wdg Run Area

Represents the watchdog driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

This should be set to RAM when other Wdg dependencies, such as Dem and Gpt, are also running from RAM. Otherwise, ROM should be selected.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ROM
literals	['RAM', 'ROM']

4.1.17 Parameter WdgCallbackNotification

Vendor specific:

WdgCallbackNotification

Callback notification for the ISR Wdg_Ipw_Isr function

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	NULL_PTR

4.1.18 Parameter WdgVersionInfoApi

Wdg VersionInfo Api

Compile switch to enable or disable the version information API.

True: API enabled

False: API disabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.1.19 Reference WdgEcucPartitionRef

Maps the Wdg driver to zero or one ECUC partition to make the driver API available in this partition.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.20 Container AutosarExt

Autosar Extension settings.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.21 Parameter WdgDisableDemReportErrorStatus

Enable/Disable Dem error reporting.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.22 Container WdgClockReferencePoint

This container contains a parameter, which represents a reference to a container of the type McuClockReferencePoint (defined in module MCU).

If the cip is trimmed (internal RC oscillator clock FIRC running at 48MHz frequency), then configure in MCU a reference point of FIRC type

Configuration Parameters

with 48MHz frequency. If the cip is not trimmed (FIRC running at frequency different than 48MHz), then configure in MCU a reference point

of CUSTOM type with the real FIRC frequency measured on the cip.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.23 Reference WdgClockReference

Reference to a container of the type *McuClockReferencePoint*, to select an input clock.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSetting↵ Config/McuClockReferencePoint

4.1.24 Container WdgSettingsConfig

WdgSettingsConfig

Configuration items for the different watchdog settings, including those for external watchdog hardware.

Included subcontainers:

- [WdgExternalConfiguration](#)
- [WdgSettingsFast](#)
- [WdgSettingsOff](#)
- [WdgSettingsSlow](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.25 Parameter WdgInstance

Vendor specific:

Wdg Hardware Instance

Select specific hardware instance for watchdog driver initialization.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	WDOG
literals	['WDOG']

4.1.26 Parameter WdgDefaultMode

Wdg Default Mode

Default mode for watchdog driver initialization.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE

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Property	Value
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	WDGIF_SLOW_MODE
literals	['WDGIF_FAST_MODE', 'WDGIF_OFF_MODE', 'WDGIF_SLOW_MODE']

4.1.27 Parameter WdgInterruptContentEnable

Vendor specific:

Wdg Interrupt Content Enable

This parameter is used to generate interrupt content for each Wdg Instance.

True = Interrupt content is generated.

False = Interrupt content is not generated.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.1.28 Reference WdgExternalTriggerCounterRef

Vendor specific:

Wdg External Trigger Counter

Reference to the GptChannel configuration which set for the watchdog servicing routine implementation.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Gpt/GptChannelConfigSet/GptChannelConfiguration']

4.1.29 Container WdgExternalConfiguration

WdgExternalConfiguration

Configuration items for an external watchdog hardware

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.30 Reference WdgExternalContainerRef

WdgExternalContainerRef

Reference to either

- a *DioChannelGroup* container in case the hardware watchdog is connected via DIO pins

- a *SpiSequenceConfiguration* container in case the watchdog hardware is accessed via SPI

Note: This parameter is not used by current implementation

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Dio/DioConfig/DioPort/DioChannelGroup', '/AUTOSAR/EcucDefs/Spi/SpiDriver/SpiSequence']

4.1.31 Container WdgSettingsFast

WdgSettingsFast

Hardware dependent settings for the watchdog driver's fast mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.32 Parameter WdgClockValue

Vendor specific:

Wdg Clock Value

This is the Implementation Specific parameter.

Indicates Wdg Clock Value in KHz (internal oscillator clock value is by default 128KHz).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0
max	56000
min	0

4.1.33 Parameter WdgRunsInStopMode

Vendor specific:

Wdg Runs in Stop Mode

This is the Implementation Specific parameter.

Enabled: Wdg continues to count even while the processor core is in stop mode.

Disabled: Wdg stops counting if the processor core is in stop mode.

Note: The 'WdgRunsInStopMode' parameter specifies if the watchdog timer should run or not while the clock to the core is halted.

This is true only for the STOP0 mode of the controller. It will always run while the controller is in the HALT0 mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.34 Parameter WdgRunsInDebugMode

Vendor specific:

Wdg Runs In Debug Mode

This is the Implementation Specific parameter.

Enabled: Wdg continues to count even while the device enters the debug mode.

Disabled: Wdg stops counting if the processor core when the device enters the debug mode

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.35 Parameter WdgRunsInWaitMode

Vendor specific:

Wdg Runs In Wait Mode

Configuration Parameters

This is the Implementation Specific parameter.

Enabled: Watchdog continues to count even while the device enters Wait mode.

Disabled: Watchdog stops counting if the processor core when the device enters Wait mode

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.36 Parameter WdgOperationMode

Vendor specific:

Wdg Operation Mode

This is the Implementation Specific parameter.

ResetOnTimeOut: Generate a reset on a time-out.

Interrupt: Generate an interrupt on an initial time-out.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	ResetOnTimeOut
literals	['ResetOnTimeOut', 'Interrupt']

4.1.37 Parameter WdgClockSelection

Vendor specific:

*Wdg Clock Selection**WDOG clock selection.*

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Bus_Clock
literals	['Bus_Clock', 'LPO_Clock', 'SOSC_Clock', 'SIRC_Clock']

4.1.38 Parameter WdgTimeoutPeriod*Vendor specific:**Wdg Timeout Period**This is the Implementation Specific parameter.Wdg Time-Out Period in seconds. Selects the time-out period for the Wdg.*

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.5
max	16776.96
min	0.0

4.1.39 Parameter WdgWindowMode*Vendor specific:**Wdg WindowMode.**Disabled: Regular mode, service sequence can be done at any time.*

Configuration Parameters

Enabled : Windowed mode, the service sequence is only valid when performed during the time frame specified by WdgWindowPeriod.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.40 Parameter WdgWindowPeriod

Vendor specific:

Wdg Window Period

This is the Implementation Specific parameter.

Wdg Window Value - When WdgWindowMode is enabled, the Wdg counter can be refreshed in the last period of the counter specified by WdgWindowPeriod.

For example, if WdgTimeoutPeriod is 0.4s and WdgWindowPeriod is 0.1s, the Wdg can be refreshed in the last 0.1s before the counter reaches 0.4s.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	16776.96
min	0.0

4.1.41 Parameter WdgPrescalerEnabled

Vendor specific:

Wdg Prescaler Enabled.

Disabled: Prescaler Disabled.

Enabled : Prescaler Enabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.42 Parameter WdgAllowUpdates

Vendor specific:

Wdg Allow Updates.

Disabled: Updates not allowed.

Enabled : Updates allowed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.43 Reference WdgClkSrcRef

Reference to the WdgClockReferencePoint from which the clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D2M30I0R0/Wdg/WdgClockReferencePoint

4.1.44 Container WdgSettingsOff

WdgSettingsOff

Hardware dependent settings for the watchdog driver's off mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.45 Parameter WdgAllowUpdates

Vendor specific:

Wdg Allow Updates.

Disabled: Updates not allowed.

Enabled : Updates allowed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.46 Container WdgSettingsSlow

WdgSettingsSlow

Hardware dependent settings for the watchdog driver's slow mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.47 Parameter WdgClockValue

Vendor specific:

Wdg Clock Value

This is the Implementation Specific parameter.

Indicates Wdg Clock Value in KHz (internal oscillator clock value is by default 128KHz).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	56000
min	0

4.1.48 Parameter WdgRunsInStopMode

Vendor specific:

Wdg Runs in Stop Mode

This is the Implementation Specific parameter.

Enabled: Wdg continues to count even while the processor core is in stop mode.

Disabled: Wdg stops counting if the processor core is in stop mode.

Note: The 'WdgRunsInStopMode' parameter specifies if the watchdog timer should run or not while the clock to the core is halted.

This is true only for the STOP0 mode of the controller. It will always run while the controller is in the HALT0 mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.49 Parameter WdgRunsInDebugMode

Vendor specific:

Wdg Runs In Debug Mode

This is the Implementation Specific parameter.

Enabled: Wdg continues to count even while the device enters the debug mode.

Disabled: Wdg stops counting if the processor core when the device enters the debug mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.50 Parameter WdgRunsInWaitMode

Vendor specific:

Wdg Runs In Wait Mode

This is the Implementation Specific parameter.

Enabled: Watchdog continues to count even while the device enters Wait mode.

Disabled: Watchdog stops counting if the processor core when the device enters Wait mode

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.51 Parameter WdgOperationMode

Vendor specific:

Wdg Operation Mode

This is the Implementation Specific parameter.

ResetOnTimeOut: Generate a reset on a time-out.

Interrupt: Generate an interrupt on an initial time-out.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ResetOnTimeOut
literals	['ResetOnTimeOut', 'Interrupt']

4.1.52 Parameter WdgClockSelection

Vendor specific:

Wdg Clock Selection

WDOG clock selection.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Bus_Clock
literals	['Bus_Clock', 'LPO_Clock', 'SOSC_Clock', 'SIRC_Clock']

4.1.53 Parameter WdgTimeoutPeriod

Vendor specific:

Wdg Timeout Period

This is the Implementation Specific parameter.Wdg Time-Out Period in seconds. Selects the time-out period for the Wdg.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
	VARIANT-LINK-TIME: LINK
valueConfigClasses	

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0.5
max	16776.96
min	0.0

4.1.54 Parameter WdgWindowMode

Vendor specific:

Wdg WindowMode.

Disabled: Regular mode, service sequence can be done at any time.

Enabled : Windowed mode, the service sequence is only valid when performed during the time frame specified by WdgWindowPeriod.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.55 Parameter WdgWindowPeriod

Vendor specific:

Wdg Window Period

This is the Implementation Specific parameter.

Wdg Window Value - When WdgWindowMode is enabled, the Wdg counter can be refreshed in the last period of the counter specified by WdgWindowPeriod.

For example, if WdgTimeoutPeriod is 0.4s and WdgWindowPeriod is 0.1s, the Wdg can be refreshed in the last 0.1s before the counter

reaches 0.4s.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	16776.96
min	0.0

4.1.56 Parameter WdgPrescalerEnabled

Vendor specific:

Wdg Prescaler Enabled.

Disabled: Prescaler Disabled.

Enabled : Prescaler Enabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.57 Parameter WdgAllowUpdates

Vendor specific:

Wdg Allow Updates.

Disabled: Updates not allowed.

Enabled : Updates allowed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.1.58 Reference WdgClkSrcRef

Reference to the *WdgClockReferencePoint* from which the clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D2M30I0R0/Wdg/WdgClockReferencePoint

4.1.59 Container WdgPublishedInformation

WdgPublishedInformation

Container holding all Wdg specific published information parameters

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.60 Parameter WdgTriggerMode

Wdg Trigger Mode

Watchdog trigger mode (toggle/window/both).

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	WDG_BOTH
literals	['WDG_BOTH', 'WDG_TOGGLE', 'WDG_WINDOW']

4.1.61 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.62 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.1.63 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.1.64 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Configuration Parameters

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.65 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	102
max	102
min	102

4.1.66 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Property	Value
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	3
max	3
min	3

4.1.67 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.68 Parameter SwPatchVersion

Vendor specific:

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0

Property	Value
max	0
min	0

4.1.69 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

`<ModuleName>_<VendorId>_<VendorApiInfix>`

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	Instance0

4.1.70 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Property	Value
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

4.2 WDOG_IP

The low level driver component can only be used with S32CT Configurator in S32 Design Studio.

- Module [Wdog_Ip](#)
 - Container [WdogUserCfg](#)
 - * Parameter [WdogDevErrorDetect](#)
 - * Parameter [WdogDisableAllowed](#)
 - * Parameter [WdogEnableTestMode](#)
 - * Parameter [WdogOsifTimeoutMethod](#)
 - * Parameter [WdogOsifTimeoutVal](#)
 - * Parameter [WdogRunArea](#)
 - * Container [WdogSettingsConfigRef](#)
 - Parameter [WdogClkSrcRef](#)
 - Parameter [WdogClockSelection](#)
 - Parameter [WdogRunsInStopMode](#)
 - Parameter [WdogRunsInDebugMode](#)
 - Parameter [WdogRunsInWaitMode](#)
 - Parameter [WdogOperationMode](#)
 - Parameter [WdogTimeoutPeriod](#)
 - Parameter [WdogWindowMode](#)
 - Parameter [WdogWindowPeriod](#)
 - Parameter [WdogPrescalerEnabled](#)
 - Parameter [WdogAllowUpdates](#)
 - Parameter [WdogCallbackNotification](#)

4.2.1 Module Wdog_Ip

Wdog config set

Included containers:

- [WdogUserCfg](#)

Configuration Parameters

Property	Value
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Property	Value
type	MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME;VARIANT-POST-BUILD;VARIANT-PRE-COMPILE

4.2.2 Container WdogUserCfg

N/A

Included subcontainers:

- [WdogSettingsConfigRef](#)

Property	Value
type	CONTAINER
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.2.3 Parameter WdogDevErrorDetect

Switch to enable / disable development error detection for this module.

True: Development error detection enabled

False: Development error detection disabled

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.2.4 Parameter WdogDisableAllowed

Switch to allow / forbid disabling the Wdog driver during runtime.

True: Disabling the Wdog driver at runtime is allowed

False: Disabling the Wdog driver at runtime is not allowed

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.2.5 Parameter WdogEnableTestMode

Switch to enable / disable a pair of APIs used to test the WDOG functionality.

True: Test Mode APIs for the Wdog driver are enabled

False: Test Mode APIs for the Wdog driver are disabled

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.2.6 Parameter WdogOsifTimeoutMethod

WdogOsifTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: Editable when 'Wdg Enable Clear Reset Request' node that require timeout are enabled.

Note: If OSIF_COUNTER_SYSTEM or OSIF_COUNTER_CUSTOM are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	OSIF_COUNTER_DUMMY
literals	[]

4.2.7 Parameter WdogOsifTimeoutVal

WdgOsifTimeoutVal.

Note: Editable when 'Wdg Enable Clear Reset Request' node that require timeout are enabled.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	3000
max	4294967295
min	0

4.2.8 Parameter WdogRunArea

Represents the watchdog driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

Property	Value
type	ENUMERATION
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	ROM
literals	[]

4.2.9 Container WdogSettingsConfigRef

N/A

Included subcontainers:

- None

Property	Value
type	CONTAINER
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.2.10 Parameter WdogClkSrcRef

The Clock configuration to be referred in Clock component

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	N/A
literals	[]

4.2.11 Parameter WdogClockSelection

Vendor specific:

Wdog Clock Selection

This is the Implementation Specific parameter.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Bus_Clock
literals	[]

4.2.12 Parameter WdogRunsInStopMode

Vendor specific:

Wdg Runs in Stop Mode

This is the Implementation Specific parameter.

Enabled: Wdg continues to count even while the processor core is in stop mode.

Disabled: Wdg stops counting if the processor core is in stop mode.

Note: The 'WdgRunsInStopMode' parameter specifies if the watchdog timer should run or not while the clock to the core is halted.

This is true only for the STOP0 mode of the controller. It will always run while the controller is in the HALT0 mode.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.2.13 Parameter WdogRunsInDebugMode

Specifies if the Wdg operates when the chip is in debug mode. If true, the Wdg counter is stopped, otherwise the Wdg counter continues to run.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
default Value	false

4.2.14 Parameter WdogRunsInWaitMode

Runs In Wait Mode.

Enabled: Watchdog continues to count even while the device enters Wait mode.

Disabled: Watchdog stops counting if the processor core when the device enters Wait mode

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.2.15 Parameter WdogOperationMode

Vendor specific:

Wdg Operation Mode

This is the Implementation Specific parameter.

ResetOnTimeOut: Generate a reset on a time-out.

Interrupt: Generate an interrupt on an initial time-out, reset on a second consecutive time-out.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ResetOnTimeOut
literals	[]

4.2.16 Parameter WdogTimeoutPeriod

Specifies the timeout value for the Wdg in clock cycles.

Property	Value
type	FLOAT
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.002
max	16776.96
min	0.0

4.2.17 Parameter WdogWindowMode

Specifies if the Wdg window mode is enabled. If true, the service sequence must be applied in the last part of the timeout period

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.2.18 Parameter WdogWindowPeriod

Wdg Window Value - When WdgWindowMode is enabled, the Wdg counter can be refreshed in the last period of the counter specified by WdgWindowPeriod.

For example, if WdgTimeoutPeriod is 0.4s and WdgWindowPeriod is 0.1s, the Wdg can be refreshed in the last 0.1s before the counter

reaches 0.4s.

Property	Value
type	FLOAT
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0.0
max	16776.96
min	0.0

4.2.19 Parameter WdogPrescalerEnabled

Wdog Prescaler Enabled.

Disabled: Prescaler Disabled

Enabled : Prescaler Enabled.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.2.20 Parameter WdogAllowUpdates

Wdog Allow Updates.

Disabled: Updates not allowed.

Enabled : Updates allowed.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.2.21 Parameter WdogCallbackNotification

Callback notification for the Wdog Isr.

Property	Value
type	FUNCTION-NAME
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	NULL_PTR

4.3 EWM_IP

The low level driver component can only be used with S32CT Configurator in S32 Design Studio.

- Module [Ewm_Ip](#)
 - Container [EwmUserCfg](#)
 - * Parameter [EwmDevErrorDetect](#)
 - * Parameter [EwmOsifTimeoutMethod](#)
 - * Parameter [EwmOsifTimeoutVal](#)
 - * Parameter [EwmRunArea](#)
 - * Container [EwmSettingsConfigRef](#)
 - Parameter [EwmClkSrcRef](#)
 - Parameter [EwmClockSelection](#)
 - Parameter [EwmTimeoutPeriod](#)
 - Parameter [EwmWindowMode](#)
 - Parameter [EwmWindowPeriod](#) _EwmWindowPeriod "EwmWindowPeriod "
 - Parameter [EwmAssertLogic](#)
 - Parameter [EwmInterruptEnable](#)
 - Parameter [EwmPrescalerValue](#)
 - Parameter [EwmCallbackNotification](#)

4.3.1 Module Ewm_Ip

Ewm config set

Included containers:

- [EwmUserCfg](#)

Property	Value
type	MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME;VARIANT-POST-BUILD;VARIANT-PRE-COMPILE

4.3.2 Container EwmUserCfg

N/A

Included subcontainers:

- [EwmSettingsConfigRef](#)

Property	Value
type	CONTAINER
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3.3 Parameter EwmDevErrorDetect

Switch to enable / disable development error detection for this module.

True: Development error detection enabled

False: Development error detection disabled

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.3.4 Parameter EwmOsifTimeoutMethod

EwmOsifTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: Editable when 'Wdg Enable Clear Reset Request' node that require timeout are enabled.

Note: If OSIF_COUNTER_SYSTEM or OSIF_COUNTER_CUSTOM are selected make sure the corresponding timer is enabled in OsIf General configuration.

Configuration Parameters

Note: Implementation Specific Parameter.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	OSIF_COUNTER_DUMMY
literals	[]

4.3.5 Parameter EwmOsifTimeoutVal

EwmOsifTimeoutVal.

Note: Editable when 'Ewm Enable Clear Reset Request' node that require timeout are enabled.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	3000
max	4294967295
min	0

4.3.6 Parameter EwmRunArea

Represents the watchdog driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

Property	Value
type	ENUMERATION
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	ROM
literals	∅

4.3.7 Container EwmSettingsConfigRef

N/A

Included subcontainers:

- None

Property	Value
type	CONTAINER
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.3.8 Parameter EwmClkSrcRef

The Clock configuration to be referred in Clock component

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1

Configuration Parameters

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	N/A
literals	[]

4.3.9 Parameter EwmClockSelection

Vendor specific:

Ewm Clock Selection

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	LPO_Clock
literals	[]

4.3.10 Parameter EwmTimeoutPeriod

Specifies the timeout value for the Ewm in second.

Property	Value
type	FLOAT
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.001
max	65.28
min	0.0

4.3.11 Parameter EwmWindowMode

Window Mode Enable.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

_EwmWindowPeriod

4.3.12 Parameter EwmWindowPeriod

Wdg Window Value - When WdgWindowMode is enabled, the Wdg counter can be refreshed in the last period of the counter specified by WdgWindowPeriod.

For example, if WdgTimeoutPeriod is 0.4s and WdgWindowPeriod is 0.1s, the Wdg can be refreshed in the last 0.1s before the counter

reaches 0.4s.

Note:For EWM, the counter is reset in other cases, refer the Reference Manual to more detail

Property	Value
type	FLOAT

Configuration Parameters

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	65.28
min	0.0

4.3.13 Parameter EwmAssertLogic

Vendor specific:

Ewm Assert Logic

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Assert_Disabled
literals	[]

4.3.14 Parameter EwmInterruptEnable

Ewm Interrupt Enable.

This is the Implementation Specific parameter.

Note: This bit when set and EWM_OUT_b is asserted, an interrupt request is generated. To de-assert interrupt request, user should clear this bit by writing 0.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.3.15 Parameter EwmPrescalerValue

Vendor specific:

Ewm Prescaler Value.

Clock prescaler value from 0 to 255.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	255
min	0

4.3.16 Parameter EwmCallbackNotification

Callback notification for the ewm isr.

Property	Value
type	FUNCTION-NAME
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	NULL_PTR

4.4 AEWDOG_IP

The low level driver component can only be used with S32CT Configurator in S32 Design Studio.

- Module [AeWdog_Ip](#)
 - Container [AeWdogUserCfg](#)
 - * Parameter [AeWdogDevErrorDetect](#)
 - * Parameter [AeWdogOsifTimeoutMethod](#)
 - * Parameter [AeWdogOsifTimeoutVal](#)
 - * Parameter [AeWdogRunArea](#)
 - * Parameter [WdgWatchdogMode](#)
 - * Parameter [WdgOperationMode](#)
 - * Parameter [WdgBadRespon](#)
 - * Parameter [AeWdogToken](#)
 - * Parameter [AeWdogEnableFaultWatchdog](#)
 - * Parameter [AeWdogFaultWatchdogWindowPeriod](#)
 - * Container [AeWdogSettingsConfigRef](#)
 - Parameter [WdgClosedWindowDutyCycle](#)
 - Parameter [WdgWindowPeriod](#)

4.4.1 Module AeWdog_Ip

AeWdog_Ip config set

Included containers:

- [AeWdogUserCfg](#)

Property	Value
type	MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME;VARIANT-POST-BUILD;VARIANT-PRE-COMPILE

4.4.2 Container AeWdogUserCfg

N/A

Included subcontainers:

- [AeWdogSettingsConfigRef](#)

Property	Value
type	CONTAINER
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.4.3 Parameter AeWdogDevErrorDetect

Switch to enable / disable development error detection for this module.

True: Development error detection enabled

False: Development error detection disabled

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.4.4 Parameter AeWdogOsifTimeoutMethod

AeWdogOsifTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: Editable when 'Wdg Enable Clear Reset Request' node that require timeout are enabled.

Note: If OSIF_COUNTER_SYSTEM or OSIF_COUNTER_CUSTOM are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	OSIF_COUNTER_DUMMY
literals	[]

4.4.5 Parameter AeWdogOsifTimeoutVal

AewdogOsifTimeoutVal.

Note: Editable when 'Aewdog Enable Clear Reset Request' node that require timeout are enabled.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	3000
max	4294967295
min	0

4.4.6 Parameter AeWdogRunArea

Represents the watchdog driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

Property	Value
type	ENUMERATION
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	ROM
literals	[]

4.4.7 Parameter WdgWatchdogMode

Watchdog mode

Configures the watchdog mode for aewdog:

CHALLENGER_WATCHDOG: Aewdog will run in Simple Mode.

SIMPLE_WATCHDOG: Aewdog will run in Challenge mode.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	SIMPLE_WATCHDOG
literals	[]

4.4.8 Parameter WdgBadRespon

Wdg max bad responses.

Configures the Max Number Of Bad Responses.

This selection is the max number of bad responses before triggering a fault response, to account for the transmission error.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	MAX_3
literals	[]

4.4.9 Parameter AeWdogToken

Wdg Watchdog Token.

Expected value in simple mode, see in challenger mode.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	23218
max	65534
min	1

4.4.10 Parameter AeWdogEnableFaultWatchdog

AeWdog Enable Fault Watchdog

This parameter is used to enable the fault watchdog.

True = Enable Fault Watchdog.

False = Disable Fault Watchdog.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.4.11 Parameter AeWdogFaultWatchdogWindowPeriod

Window Period for Fault Watchdog.

Property	Value
type	INTEGER
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Configuration Parameters

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	2
max	8000
min	2

4.4.12 Container AeWdogSettingsConfigRef

N/A

Included subcontainers:

- None

Property	Value
type	CONTAINER
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.4.13 Parameter WdgClosedWindowDutyCycle

Watchdog Closed Window Duty Cycle

Response must be sent during open window only. Otherwise, an error is triggered.

DUTY_DISABLED: the feature will be disabled.

Property	Value
type	ENUMERATION
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	DUTY_DISABLED
literals	[]



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Module Documentation

6.1 AeWdog_Ip

6.1.1 Detailed Description

Data Structures

- struct [AeWdog_Ip_ConfigType](#)

AEWDOG configuration structure This structure is used to configure the AEWDOG fault response, watchdog mode, Max number of bad responses, window duty cycle and watchdog window duration. [More...](#)

Enum Reference

- enum [AeWdog_Ip_StatusType](#)

Enum defining the possible type values for AEWDOG API @Implements AeWdog_Ip_StatusType enumeration.

- enum [AeWdog_Ip_ModeType](#)

Enum defining the watchdog mode for AEWDOG API @Implements AeWdog_Ip_ModeType enumeration.

- enum [AeWdog_Ip_FaultRespType](#)

Enum defining the fault response configuration for AEWDOG API @Implements AeWdog_Ip_FaultRespType enumeration.

- enum [AeWdog_Ip_BadRespType](#)

Enum defining the max number of bad responses for AEWDOG API @Implements AeWdog_Ip_BadRespType enumeration.

- enum [AeWdog_Ip_WindowDutyCycleType](#)

Enum defining the window duty cycle for AEWDOG API @Implements AeWdog_Ip_WindowDutyCycleType enumeration.

- enum [AeWdog_Ip_WindowPeriodType](#)

Enum defining the watchdog window duration for AEWDOG API @Implements AeWdog_Ip_WindowPeriodType enumeration.

6.1.2 Data Structure Documentation

6.1.2.1 struct AeWdog_Ip_ConfigType

AEWDOG configuration structure This structure is used to configure the AEWDOG fault response, watchdog mode, Max number of bad responses, window duty cycle and watchdog window duration.

Implements : aewdog_init_config_t_Class

Definition at line 188 of file AeWdog_Ip_Types.h.

Data Fields

AeWdog_Ip_ModeType	eAeWdogMode	Watchdog mode for AEWDOG
AeWdog_Ip_FaultRespType	eFaultRespType	Fault response configuration
AeWdog_Ip_BadRespType	eMaxBadResp	Max number of bad responses
AeWdog_Ip_WindowDutyCycleType	eWindowDutyCycle	Window duty cycle
AeWdog_Ip_WindowPeriodType	eWindowPeriod	Watchdog Window Duration

6.1.3 Enum Reference

6.1.3.1 AeWdog_Ip_StatusType

enum [AeWdog_Ip_StatusType](#)

Enum defining the possible type values for AEWDOG API @Implements AeWdog_Ip_StatusType_enumeration.

Definition at line 106 of file AeWdog_Ip_Types.h.

6.1.3.2 AeWdog_Ip_ModeType

enum [AeWdog_Ip_ModeType](#)

Enum defining the watchdog mode for AEWDOG API @Implements AeWdog_Ip_ModeType_enumeration.

Definition at line 116 of file AeWdog_Ip_Types.h.

6.1.3.3 AeWdog_Ip_FaultRespType

enum [AeWdog_Ip_FaultRespType](#)

Enum defining the fault response configuration for AEWDOG API @Implements AeWdog_Ip_FaultRespType_↔ enumeration.

Definition at line 125 of file AeWdog_Ip_Types.h.

6.1.3.4 AeWdog_Ip_BadRespType

enum [AeWdog_Ip_BadRespType](#)

Enum defining the max number of bad responses for AEWDOG API @Implements AeWdog_Ip_BadRespType_↔ enumeration.

Definition at line 134 of file AeWdog_Ip_Types.h.

6.1.3.5 AeWdog_Ip_WindowDutyCycleType

enum [AeWdog_Ip_WindowDutyCycleType](#)

Enum defining the window duty cycle for AEWDOG API @Implements AeWdog_Ip_WindowDutyCycleType_↵
enumeration.

Definition at line 145 of file AeWdog_Ip_Types.h.

6.1.3.6 AeWdog_Ip_WindowPeriodType

enum [AeWdog_Ip_WindowPeriodType](#)

Enum defining the watchdog window duration for AEWDOG API @Implements AeWdog_Ip_WindowPeriodType_↵
enumeration.

Definition at line 156 of file AeWdog_Ip_Types.h.

6.2 Ewm_Ip

6.2.1 Detailed Description

Data Structures

- struct [Ewm_Ip_ConfigType](#)

EWM configuration structure This structure is used to configure the EWM prescaler, window, interrupt and input pin. [More...](#)

Types Reference

- typedef void(* [Ewm_Ip_CallbackPtrType](#)) (void)

EWM callback type Implements : Ewm_Ip_CallbackPtrType_Class.

Enum Reference

- enum [Ewm_Ip_StatusType](#)

Enum defining the possible type values for EWM API.

- enum [Ewm_Ip_Assert_LogicType](#)

EWM input pin configuration Configures if the input pin is enabled and when is asserted.

6.2.2 Data Structure Documentation

6.2.2.1 struct Ewm_Ip_ConfigType

EWM configuration structure This structure is used to configure the EWM prescaler, window, interrupt and input pin.

Implements : ewm_init_config_t_Class

Definition at line 140 of file Ewm_Ip_Types.h.

Data Fields

Ewm_Ip_Assert_LogicType	AssertLogic	Assert logic for EWM input pin
boolean	InterruptEnable	Enable EWM interrupt
uint8	Prescaler	EWM clock prescaler
uint8	CompareLow	Compare low value
uint8	CompareHigh	Compare high value
Ewm_Ip_CallbackPtrType	pfEwmCallback	Interrupt callback

6.2.3 Types Reference

6.2.3.1 Ewm_Ip_CallbackPtrType

typedef void(* Ewm_Ip_CallbackPtrType) (void)

EWM callback type Implements : Ewm_Ip_CallbackPtrType_Class.

Definition at line 131 of file Ewm_Ip_Types.h.

6.2.4 Enum Reference

6.2.4.1 Ewm_Ip_StatusType

enum `Ewm_Ip_StatusType`

Enum defining the possible type values for EWM API.

Definition at line 104 of file Ewm_Ip_Types.h.

6.2.4.2 Ewm_Ip_Assert_LogicType

enum `Ewm_Ip_Assert_LogicType`

EWM input pin configuration Configures if the input pin is enabled and when is asserted.

Enumerator

EWM_IN_ASSERT_DISABLED	Input pin disabled
EWM_IN_ASSERT_ON_LOGIC_ZERO	Input pin asserts EWM when on logic 0
EWM_IN_ASSERT_ON_LOGIC_ONE	Input pin asserts EWM when on logic 1

Definition at line 115 of file Ewm_Ip_Types.h.

6.3 Wdg

6.3.1 Detailed Description

Data Structures

- struct [Wdg_ConfigType](#)
Defines the configuration structure. [More...](#)

Enum Reference

- enum [Wdg_ServiceIdType](#)
This enumerated type will contain the service ids for the watchdog functions.
- enum [Wdg_ErrorIdType](#)
Indicates the additional det errors used by the watchdog driver.
- enum [Wdg_Ipw_InstanceType](#)
Wdg_Ipw_InstanceType.
- enum [Wdg_Ipw_IpType](#)
Wdg_Ipw_IpType.

Function Reference

- void [Wdg_ChannelInit](#) (const [Wdg_Ipw_InstanceType](#) Instance, const [Wdg_ConfigType](#) *ConfigPtr)
This function initializes the WDG module.
- Std_ReturnType [Wdg_ChannelSetMode](#) (const [Wdg_Ipw_InstanceType](#) Instance, WdgIf_ModeType Mode)
Switches the watchdog into the mode Mode.
- void [Wdg_ChannelSetTriggerCondition](#) (const [Wdg_Ipw_InstanceType](#) Instance, uint16 Timeout)
Reset the watchdog timeout counter according to the timeout value passed.

6.3.2 Data Structure Documentation

6.3.2.1 struct Wdg_ConfigType

Defines the configuration structure.

Definition at line 207 of file Wdg_ChannelTypes.h.

Data Fields

- const WdgIf_ModeType [Wdg_DefaultMode](#)
The number of configured channels.
- const [Wdg_Ipw_InstanceType](#) [Wdg_Instance](#)
The instance id.
- const Gpt_ChannelType [Wdg_TimerChannel](#)
Gpt Channel configured.
- const uint32 [Wdg_u32TriggerSourceClock](#)
The frequency of the configured timer channel.
- const Wdg_ModeType *const [Wdg_ModeSettings](#) [3]
Pointer to Watchdog Specific implementation details.

Field Documentation

Wdg_DefaultMode

```
const WdgIf_ModeType Wdg_DefaultMode
```

The number of configured channels.

Definition at line 212 of file Wdg_ChannelTypes.h.

Wdg_Instance

```
const Wdg_Ipw_InstanceType Wdg_Instance
```

The instance id.

Definition at line 216 of file Wdg_ChannelTypes.h.

Wdg_TimerChannel

```
const Gpt_ChannelType Wdg_TimerChannel
```

Gpt Channel configured.

Definition at line 222 of file Wdg_ChannelTypes.h.

Wdg_u32TriggerSourceClock

```
const uint32 Wdg_u32TriggerSourceClock
```

The frequency of the configured timer channel.

Definition at line 227 of file Wdg_ChannelTypes.h.

Wdg_ModeSettings

```
const Wdg_ModeType* const Wdg_ModeSettings[3]
```

Pointer to Watchdog Specific implementation details.

Definition at line 233 of file Wdg_ChannelTypes.h.

6.3.3 Enum Reference

6.3.3.1 Wdg_ServiceIdType

```
enum Wdg_ServiceIdType
```

This enumerated type will contain the service ids for the watchdog functions.

Precondition

To define WDG_GETVERSION_ID, WDG_VERSION_INFO_API has to be equal to STD_ON
Definition at line 153 of file Wdg_ChannelTypes.h.

6.3.3.2 Wdg_ErrorIdType

enum [Wdg_ErrorIdType](#)

Indicates the additional det errors used by the watchdog driver.

Definition at line 177 of file Wdg_ChannelTypes.h.

6.3.3.3 Wdg_Ipw_InstanceType

enum [Wdg_Ipw_InstanceType](#)

Wdg_Ipw_InstanceType.

Contains the information related to available Wdg Instances.

Definition at line 177 of file Wdg_Ipw_Types.h.

6.3.3.4 Wdg_Ipw_IpType

enum [Wdg_Ipw_IpType](#)

Wdg_Ipw_IpType.

Contains the Ip types available for Wdg.

Definition at line 188 of file Wdg_Ipw_Types.h.

6.3.4 Function Reference

6.3.4.1 Wdg_ChannelInit()

```
void Wdg_ChannelInit (
    const Wdg\_Ipw\_InstanceType Instance,
    const Wdg\_ConfigType * ConfigPtr )
```

This function initializes the WDG module.

The Wdg_Init function shall initialize the Wdg module and the watchdog hardware, i.e. it shall set the default watchdog mode and timeout period as provided in the configuration set.

Parameters

in	<i>ConfigPtr</i>	Pointer to configuration set.
in	<i>Instance</i>	Harwdware instance.

Returns

void

6.3.4.2 Wdg_ChannelSetMode()

```
Std_ReturnType Wdg_ChannelSetMode (
    const Wdg_Ipw_InstanceType Instance,
    WdgIf_ModeType Mode )
```

Switches the watchdog into the mode Mode.

By choosing one of a limited number of statically configured settings (e.g. toggle or window watchdog, different timeout periods) the Wdg module and the watchdog hardware can be switched between the following three different watchdog modes using the Wdg_SetMode function:

- WDGIF_OFF_MODE,
- WDGIF_SLOW_MODE,
- WDGIF_FAST_MODE.

Parameters

in	<i>Mode</i>	One of the following statically configured modes: 1. WDGIF_OFF_MODE, 2. WDGIF_SLOW_MODE, 3. WDGIF_FAST_MODE.
in	<i>Instance</i>	Harwdware instance.

Returns

Std_ReturnType.

Return values

<i>E_OK</i>	Mode switch executed completely and successfully.
<i>E_NOT_OK</i>	The mode switch encountered errors.

6.3.4.3 Wdg_ChannelSetTriggerCondition()

```
void Wdg_ChannelSetTriggerCondition (
    const Wdg_Ipw_InstanceType Instance,
    uint16 Timeout )
```

Reset the watchdog timeout counter according to the timeout value passed.

Parameters

in	<i>Timeout</i>	value (milliseconds) for setting the trigger counter.
in	<i>Instance</i>	Harwdware instance.

6.4 Wdog_Ip

6.4.1 Detailed Description

Data Structures

- struct [Wdog_Ip_OpModeType](#)

WDOG option mode configuration structure Implements : Wdog_Ip_OpModeType_Class. [More...](#)

- struct [Wdog_Ip_ConfigType](#)

WDOG user configuration structure Implements : Wdog_Ip_ConfigType_Class. [More...](#)

Types Reference

- typedef void(* [Wdog_Ip_CallbackPtrType](#)) (void)

WDOG callback type Implements : Wdog_Ip_CallbackPtrType_Class.

Enum Reference

- enum [Wdog_Ip_StatusType](#)

Enum defining the possible type values for WDOG API @Implements Wdog_Ip_StatusType_enumeration.

- enum [Wdog_Ip_ClkSourceType](#)

Clock sources for the WDOG. Implements : Wdog_Ip_ClkSourceType_Class.

6.4.2 Data Structure Documentation

6.4.2.1 struct Wdog_Ip_OpModeType

WDOG option mode configuration structure Implements : Wdog_Ip_OpModeType_Class.

Definition at line 153 of file Wdog_Ip_Types.h.

Data Fields

boolean	bWait	Wait mode
boolean	bStop	Stop mode
boolean	bDebug	Debug mode

6.4.2.2 struct Wdog_Ip_ConfigType

WDOG user configuration structure Implements : Wdog_Ip_ConfigType_Class.

Definition at line 164 of file Wdog_Ip_Types.h.

Data Fields

Wdog_Ip_ClkSourceType	clkSource	The clock source of the WDOG
Wdog_Ip_OpModeType	opMode	The modes in which the WDOG is functional
boolean	UpdateEnable	If true, further updates of the WDOG are enabled
boolean	IntEnable	If true, an interrupt request is generated before reset
boolean	WinEnable	If true, window mode is enabled
uint16	WindowValue	The window value
uint16	TimeoutValue	The timeout value

Data Fields

boolean	PrescalerEnable	If true, a fixed 256 prescaling of the counter reference clock is enabled
Wdog_Ip_CallbackPtrType	pfWdogCallback	Interrupt callback

6.4.3 Types Reference

6.4.3.1 Wdog_Ip_CallbackPtrType

```
typedef void(* Wdog_Ip_CallbackPtrType) (void)
```

WDOG callback type Implements : Wdog_Ip_CallbackPtrType_Class.

Definition at line 147 of file Wdog_Ip_Types.h.

6.4.4 Enum Reference

6.4.4.1 Wdog_Ip_StatusType

```
enum Wdog_Ip_StatusType
```

Enum defining the possible type values for WDOG API @Implements Wdog_Ip_StatusType_enumeration.

Definition at line 105 of file Wdog_Ip_Types.h.

6.4.4.2 Wdog_Ip_ClkSourceType

```
enum Wdog_Ip_ClkSourceType
```

Clock sources for the WDOG. Implements : Wdog_Ip_ClkSourceType_Class.

Enumerator

WDOG_IP_BUS_CLOCK	Bus clock
WDOG_IP_LPO_CLOCK	LPO clock
WDOG_IP_SOSC_CLOCK	SOSC clock
WDOG_IP_SIRC_CLOCK	SIRC clock

Definition at line 116 of file Wdog_Ip_Types.h.

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